2.3 Machine Learning vs. Deep Learning

Introduction

In machine learning (ML), people usually choose the important details (features) by hand.

In deep learning (DL), a neural network finds those details on its own. Below is one example of each.

Example 1: Machine Learning – House price Predicition

Real-world application: Estimated home prices are displayed on applications such as Redfin and Zillow. Clear details like bedrooms, square footage and neighborhood ratings can be used in a straightforward model like linear regression to do this. It predicts the cost of a new house after learning from previous sales.

Why ML: People know of the particulars that affect price, and the data is arranged neatly in row and columns. Because you can illustrate how each detail affects the pricing, the model is simple to explain, requires minimal data, and trains quickly.

Why DL doesn't work: A deep network has little to find in a brief set of precise details. It requires a lot of data and processing power, and it is more difficult to justify a home's price.

Example 2: Deep Learning – Natural Language Processing

Real -world application: At work, I worked on the AI hiring software Hireaide. We used Hugging Face Transformers and pre-trained SBERT to match resumes with job descriptions. Although "built web apps with React" and "front-end developer" are two separate terms, a resume that uses the former might be suitable for the latter title. This resulted in a 20% decrease in manual hiring.

Why DL fits*:* People describe the same skill in different ways, and resumes are messy texts. We didn't require a list of every keyword because a deep model learns meaning from context.

Why ML doesn’t work: A basic model needs elements, such as keywords. It couldn't distinguish between "Java" the language and "Java" the island, and it would overlook different wording for the same talent.

References:

https://huggingface.co/docs/transformers/index

https://aimultiple.com/deep-learning-applications

https://aiml.com/